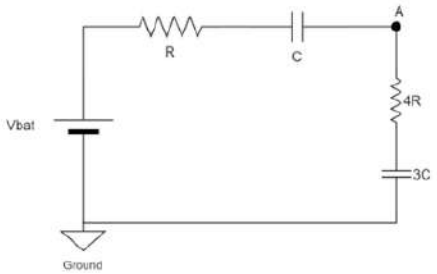
Question 1

Revisit Later

Consider the circuit below. Suppose Vbat is 20V, R is 5 ohm and C is 10 mF, what will be the steady state voltage at node A?



Select an option

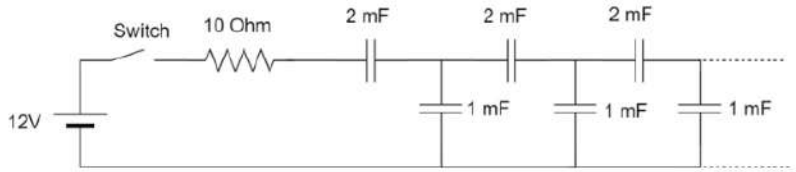
Clear Response

- 0V
- 4V
- 5V
- 15V
- 16V
- 20V

Question 2

Revisit Later

Consider the hypothetical circuit below, formed by an infinite ladder of 1 mF and 2 mF capacitors. Suppose all capacitors are initially discharged and at time t = 0, the switch is turned ON. What is (roughly) the amount of charge that will pass through the 10 ohm resistor in a duration of 1 seconds?



Select an option

Clear Response

- 0 mC
- 3 mC
- 4 mC
- 8 mC
- 12 mC
- Infinite

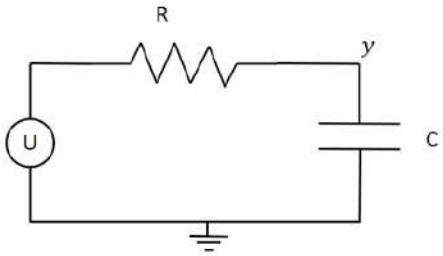
Next

Question 3

Revisit Later

Suppose in the circuit below, the voltage of the supply U is updated with the intention to maintain the value across capacitor as y_0. To achieve this target, a Proportional regulator (i..e only the P regulator of the PID regulator) is used with the control law as below, where y is the instantaneous voltage across the capacitor

$$U = K \times (y_0 - y)$$



What will be the steady state voltage value at node y?

Select an option

Clear Response

- y_0
- $y_0 \times \frac{k}{k+1}$
- $y_0 \times \frac{2k}{2k+1}$
- $0V$
- Depends on values of R and C

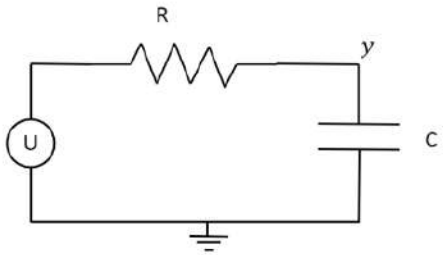
Next

Question 4

Revisit Later

Suppose in the circuit below, the voltage of the supply U is updated with the intention to maintain the value across capacitor as y_0. To achieve this target, a Proportional regulator (i..e only the P regulator of the PID regulator) is used with the control law as below, where y is the instantaneous voltage across the capacitor

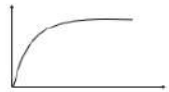
$$U = K \times (y_0 - y)$$



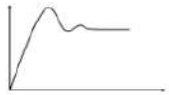
What will be qualitative nature of waveform across the capacitor?

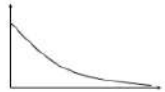
Select an option

Clear Response







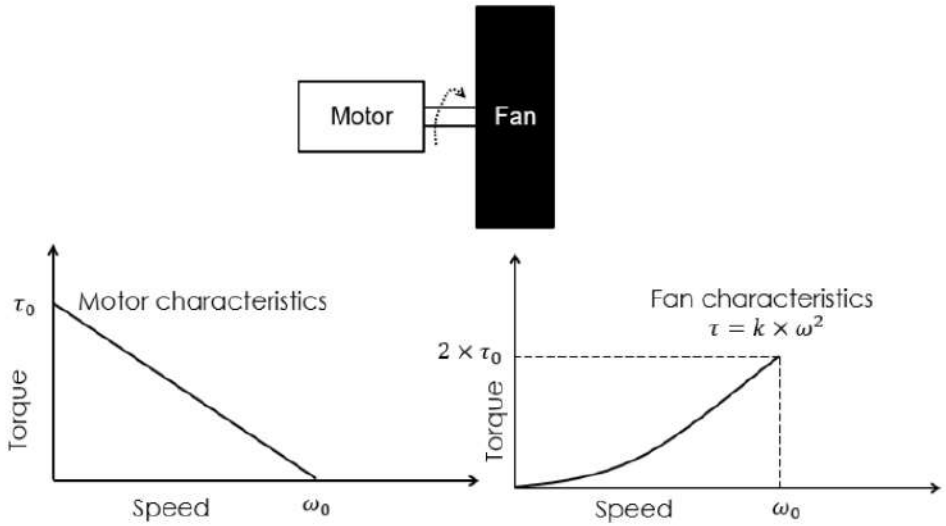


Next

Question 5

Revisit Later

Suppose a motor is driving a fan. The motor's torque-vs-speed characteristic is a straight line as shown in the figure. The fan's load torque as a function of speed is has a quadratic relationship, as shown figure.
What will be steady state operating speed of the motor?



Select an option

Clear Response

- $\omega_0/4$
- $\omega_0/3$
- $\omega_0/2$
- $2\omega_0/3$
- $3\omega_0/4$
- ω

Next

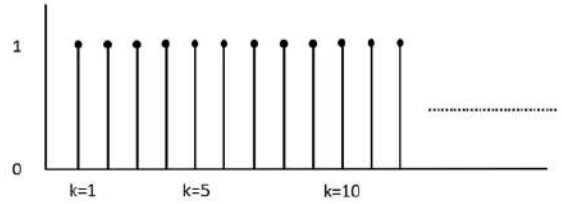
Question 6

Revisit Later

Suppose a micro-controller based system is sensing an input signal on it's analog-to-digital converter and implementing a filter as below, with $y_0 = 0$ as the initial condition.

$$y_k = \alpha x_k + \beta y_{k-1}$$

Suppose the input samples are of constant value, as shown in the figure below. Also Suppose $\alpha = 1$ And $\beta = 0.5$. What will be the steady state value of y for large values of k?



Select an option

Clear Response

- 0.5
- 1
- 1.5
- 2

Next

Question 7

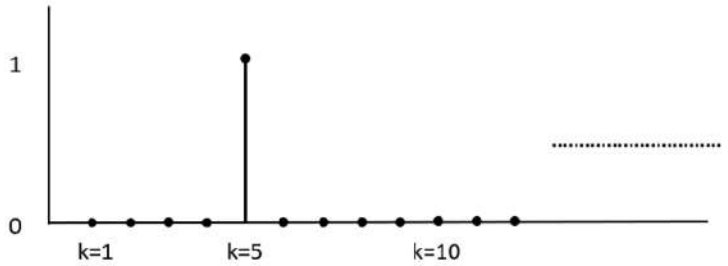
Revisit Later

Suppose a micro-controller based system is sensing an input signal on it's analog-to-digital converter and implementing a filter as below, with $y_0 = 0$ as the initial condition.

$$y_k = \alpha x_k + \beta y_{k-1}$$

Suppose the input is of "impulse" form, as shown in the figure below.

Also Suppose $\alpha = 0.5$ And $\beta = 2$. What will be the value of y for $k = 10$?



Select an option

Clear Response

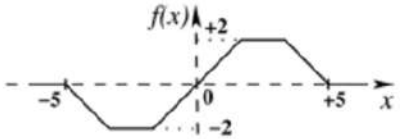
- 16
- 12
- 8
- 4

Next

Question 8

Revisit Later

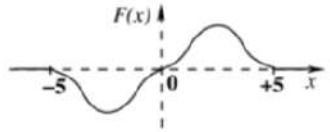
Consider the plot of $f(x)$ verses x as shown below.

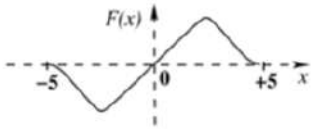


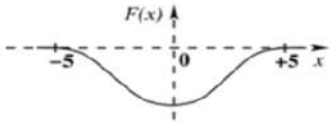
Suppose $F(x) = \int_{-5}^{x} f(y)dy$. Which one of the following is a graph of $F(x)$

Select an option

Clear Response







Next

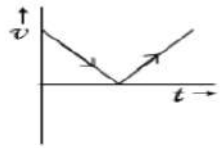
Question 9

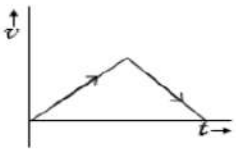
Revisit Later

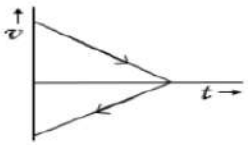
A body is thrown vertically upwards. Which one of the following graphs correctly represents the velocity vs time?

Select an option

Clear Response





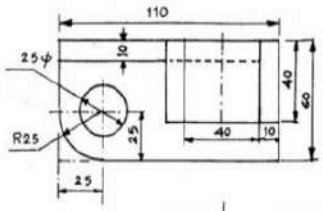


Next

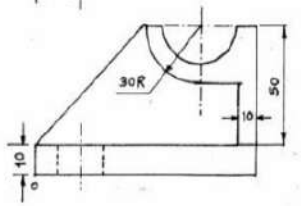
Question 10

Revisit Later

Choose the correct isometric view of the part whose top and front view are shown below?



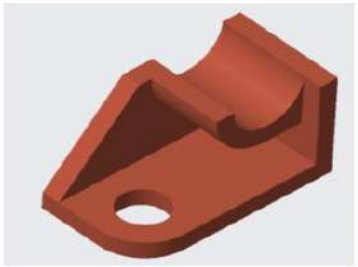
Top View

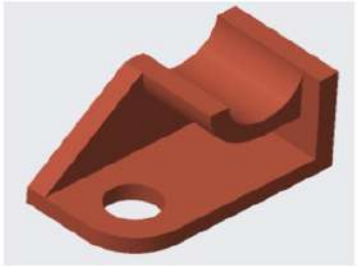


Front View

Select an option

Clear Response



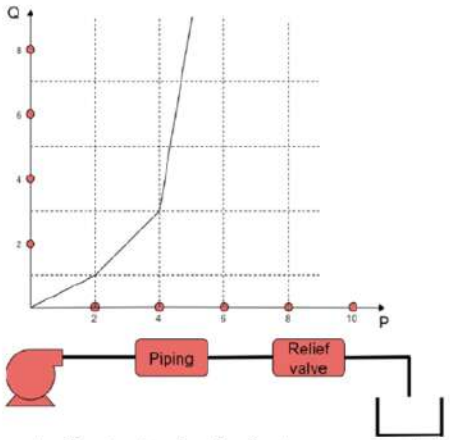


Next

Question 11

Revisit Later

Consider a hydraulic device, called a Relief valve whose pressure P - flow Q characteristic is as shown below. This is typical of pressure relief valves which do not allow the pressure drop to exceed a certain limit.



Suppose the relief valve is connected in a hydraulic circuit where a pump delivers fluid at a constant pressure of 10 bar above atmospheric pressure. The piping system is assumed to have a pressure - flow relationship given by **Q = P/2** Assume that the relief valve discharges the fluid into a tank which is at atmospheric pressure (1 bar). What value will the steady state pressure drop across the relief valve have?

Select an option

Clear Response

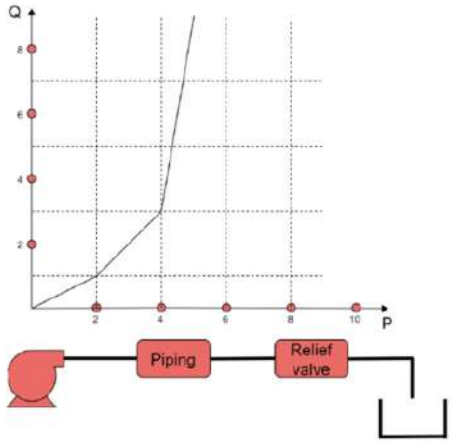
- 2
- 4
- 6
- 9

Next

Question 12

Revisit Later

Consider a hydraulic device, called a Relief valve whose pressure P - flow Q characteristic is as shown below. This is typical of pressure relief valves which do not allow the pressure drop to exceed a certain limit.



Now suppose the piping section above is replaced by an orifice whose pressure - flow relationship is given by $\mathbf{Q} = \sqrt{P/8}$. What will the new value for steady state pressure drop across the relief valve be?

Select an option

Clear Response

- 2
- 4
- 6
- 9

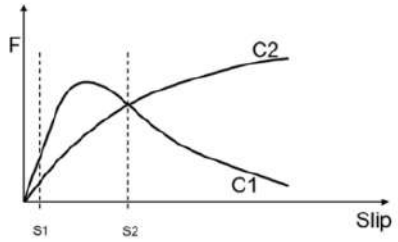
Next

Question 13

Revisit Later

The graphs C1 and C2 in below figure shows the amount of braking (retarding force **F**) as a function of wheel slip for a vehicle cruising along a road in two different roads. Slip is defined as **(v_vehicle - v_wheel_speed)/v_vehicle** expressed as a percentage. Here **v_vehicle** is the true speed of the vehicle while **v_wheel_speed** is the speed as computed from the rotational speed of the wheels. Suppose we know that increased braking by the driver (i.e, as the driver brakes harder) increases slip.

In road 1 where the graph C1 is applicable, the driver finds him/her-self in two situations. In one case the vehicle slip is **S1** and in the other case the slip is **S2**. The driver wants to stop the vehicle in as short a distance as possible. In both cases, the initial braking applied by the driver is about 25% of the maximum braking possible. In which situation should s/he brake harder



Select an option

Clear Response

- S1 and S2
- S1 but not S2
- S2 but not S1
- Neither S1 nor S2

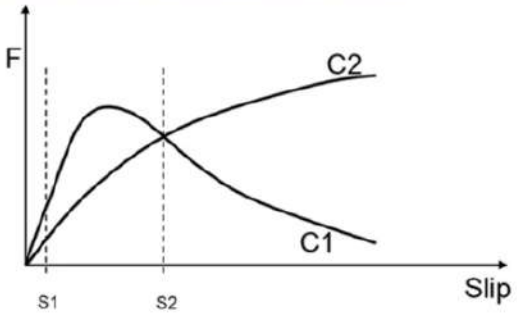
Next

Question 14

Revisit Later

The graphs C1 and C2 in below figure shows the amount of braking (retarding force **F**) as a function of wheel slip for a vehicle cruising along a road in two different roads. Slip is defined as **(v_vehicle - v_wheel_speed)/v_vehicle** expressed as a percentage. Here **v_vehicle** is the true speed of the vehicle while **v_wheel_speed** is the speed as computed from the rotational speed of the wheels. Suppose we know that increased braking by the driver (i.e, as the driver brakes harder) increases slip.

Now suppose the tire and road materials are such that graph C2 is applicable. Once again, in which situation should s/he brake harder?



Select an option

Clear Response

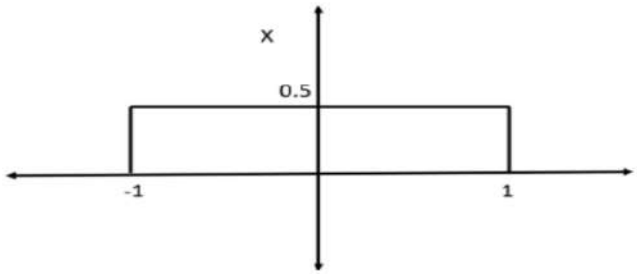
- S1 and S2
- S1 but not S2
- S2 but not S1
- Neither S1 nor S2

Next

Question 15

Revisit Later

Consider a function random(), which generates a random number with a uniform probability distribution between -1 and +1 as shown in figure below.



Consider the below program in a pseudo language with a C-like syntax (consider all variables as floating-point numbers). What will be the (approximate) output expected from the program?

```
Y = 0
for (a = 0; a < 1000000; a++)
{
   X1 = random()
   X2 = random()
   Y = Y + X1*X2
}
Y = Y/1000000
print(Y)
```

Select an option

Clear Response

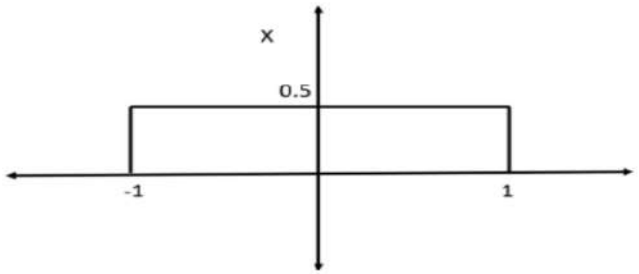
- 0
- 1/4
- 1/3
- 1/2

Next

Question 16

Revisit Later

Consider a function random(), which generates a random number with a uniform probability distribution between -1 and +1 as shown in figure below.



Consider another program. What will be the (approximate) output expected from this program?

```
Y = 0
for (a = 0; a < 1000000; a++)
{
    X = random()
    Y = Y + X*X
}
Y = Y/1000000
print(Y)
```

Select an option

Clear Response

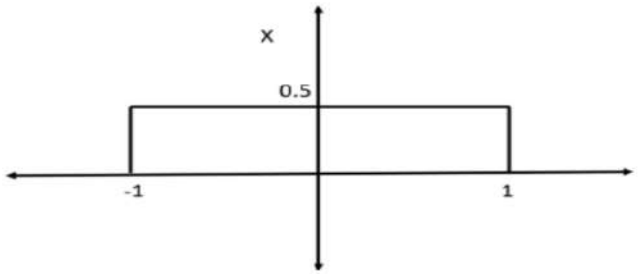
- 0
- 1/4
- 1/3
- 1/2

Next

Question 17

Revisit Later

Consider a function random(), which generates a random number with a uniform probability distribution between -1 and +1 as shown in figure below.



Consider another program. What will be the (approximate) output expected from this program?

```
function abs(x) {
   If x > 0
      return x
   else
      return -x
}


Y = 0
for (a = 0; a < 1000000; a++)
{
```

Select an option

Clear Response

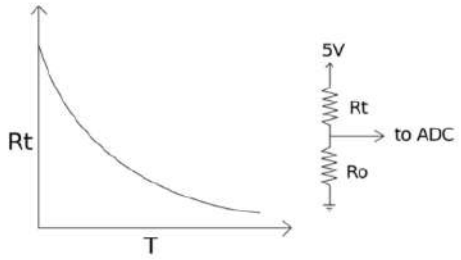
- 0
- 1/4
- 1/3
- 1/2

Next

Question 18

Revisit Later

Thermistors are typically used as temperature sensors by leveraging their temperature-dependent-resistance property. Below figure shows the qualitative change in resistance of an example thermistor, as a function of temperature. One method of interfacing thermistors with microcontrollers for sensor measurement is to use a resistor divider, where **Rt** is the (variable) resistance of the thermistor, while **Ro** is some fixed resistance. The ADC counts read by microcontroller change as temperature changes.



Suppose we are using an ADC with infinitely good resolution. In which temperature range will you prefer operating the thermistor to have greater confidence in the sensor measurement readings (purely because of temperature vs resistance characteristics of the thermistor) ?

Select an option

Clear Response

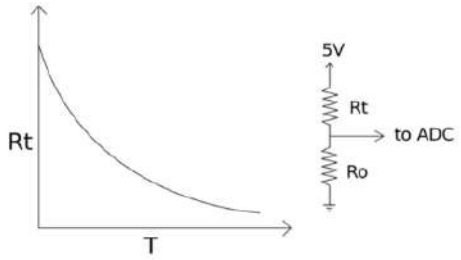
- At high temperatures
- At low temperatures
- No such preference.
- Confidence on temperature sensor readings should be similar at all temperature ranges

Next

Question 19

Revisit Later

Thermistors are typically used as temperature sensors by leveraging their temperature-dependent-resistance property. Below figure shows the qualitative change in resistance of an example thermistor, as a function of temperature. One method of interfacing thermistors with microcontrollers for sensor measurement is to use a resistor divider, where **Rt** is the (variable) resistance of the thermistor, while **Ro** is some fixed resistance. The ADC counts read by microcontroller change as temperature changes.



Assuming Ro = 1 kohm and Rt = 4 kohm at 25 degC and Rt = 2 kohm at 50 degC with a fairly linear characteristic between 25 degC and 50 degC, what is the estimated temperature if ADC counts is measured as 256 with 10 bit resolution between 0 - 5V?

Select an option

Clear Response

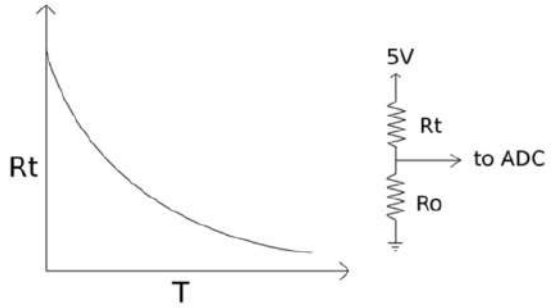
- 30 degC
- 32.5 degC
- 35 degC
- 37.5 degC
- 40 degC

Next

Question 20

Revisit Later

Thermistors are typically used as temperature sensors by leveraging their temperature-dependent-resistance property. Below figure shows the qualitative change in resistance of an example thermistor, as a function of temperature. One method of interfacing thermistors with microcontrollers for sensor measurement is to use a resistor divider, where **Rt** is the (variable) resistance of the thermistor, while **Ro** is some fixed resistance. The ADC counts read by microcontroller change as temperature changes.



In reference to Question 19, if **Ro** in the circuit is replaced with **Rt**, such that the two thermistors are effectively in series configuration, with both thermistors reading temperature of the same object. How would the equivalent temperature measurement resolution change in with respect to previous configuration mentioned in Question 19 ?

Select an option

Clear Response

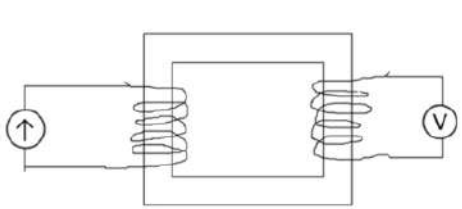
- Resolution increases
- Resolution remains same
- Resolution reduces
- Not enough information
- ADC counts no longer have any correlation with temperature

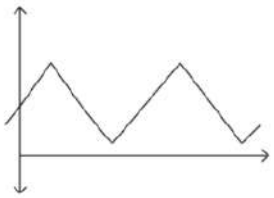
Next

Question 21

Revisit Later

A transformer with 1:1 turns ratio is shown. The primary side is powered by a current signal generator, while the secondary side voltage is monitored by an oscilloscope. We can assume ideal transformer with no leakage flux and no core saturation. Say the current source is such that it pumps in current as shown in adjacent figure. How will the voltage measured at output look like?





Select an option

Clear Response

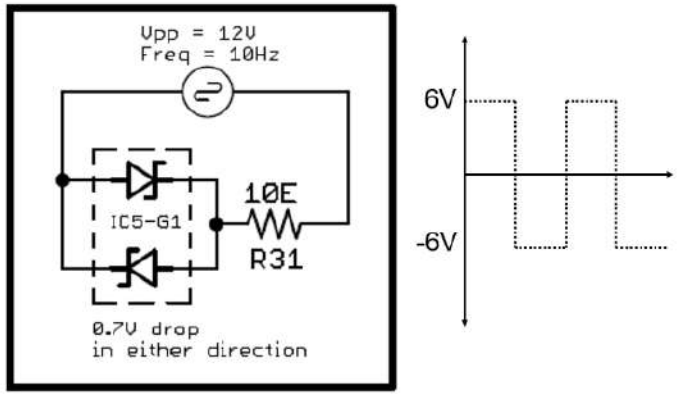
- Square waveform with zero average voltage
- Square waveform with positive average voltage
- Square waveform with negative average voltage
- Triangular waveform with zero average voltage
- Triangular waveform with positive average voltage

Next

Question 22

Revisit Later

In the circuit shown, imagine that ideal diodes with forward drop voltage of 0.7V, and 100V breakdown voltage in reverse direction are used. In parallel (as shown). The waveform of voltage source with peak-to-peak voltage of 12V is shown in the adjacent graph. What would be the heat dissipation in the unit IC5 in the circuit shown?



Select an option

Clear Response

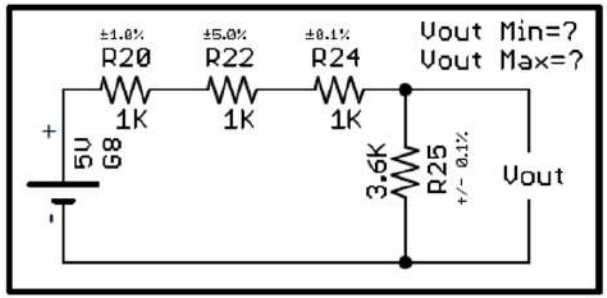
- 371 mW
- 840 mW
- 565 mW
- 791 mW

Next

Question 23

Revisit Later

Consider a voltage divider circuitry as shown circuitry. The tolerance of each resistor is mentioned in the circuit schematic. What would be the maximum value of Vout in this circuitry?



Select an option

Clear Response

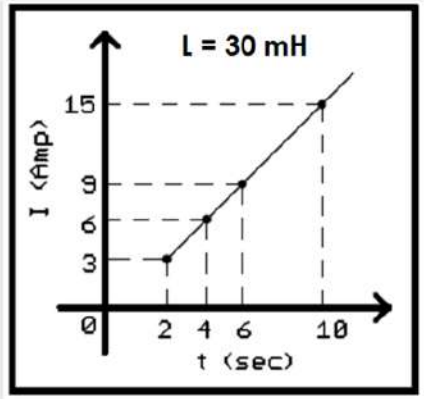
- 2727 mV
- 2743 mV
- 2701 mV
- 2754 mV
- 2783 mV

Next

Question 24

Revisit Later

Consider an inductor through which current is flown as per shown graph. What is value of voltage induced across the inductor?



Select an option

Clear Response

- 15 mV
- 2 mV
- 20 mV
- 45 mV

Next

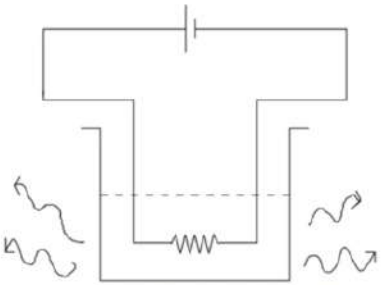
Question 25

Revisit Later

A resistor powered by a battery is immersed in metallic oil tub as shown in figure below. The ambient temperature around the system is maintained at a steady 25 degC. Thermal radiation of the resistor and oil tub combination has been empirically approximated using the formula

$$\text{Power dissipated} = 4(T - T_a)^2$$

Where T_a is the ambient temperature (25 degC). It was further observed that the system radiates heat from all sides (ie, from oil surface, as well as from metallic oil tub)



With a battery voltage of 10V and resistor value of 1 ohm, estimate the steady state temperature of the oil tub.

Select an option

Clear Response

- 26 degC
- 28 degC
- 30 degC
- 32 degC
- 34 degC

Next

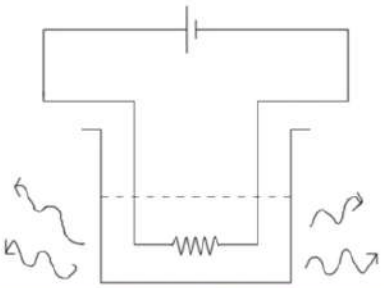
Question 26

Revisit Later

A resistor powered by a battery is immersed in metallic oil tub as shown in figure below. The ambient temperature around the system is maintained at a steady 25 degC. Thermal radiation of the resistor and oil tub combination has been empirically approximated using the formula

$$\text{Power dissipated} = 4(T - T_a)^2$$

Where T_a is the ambient temperature (25 degC). It was further observed that the system radiates heat from all sides (ie, from oil surface, as well as from metallic oil tub)



In continuation of above, experimental results suggested that the estimated value of temperature does not match the observed value as thermal coefficient of resistance was not accounted. The experimental steady state temperature was observed as 29 degC. Assuming that the resistor value was 1 ohm at 25 degC, what is the likely "average" value of thermal coefficient of resistor between 25 degC and 29 degC? Resistor value variation with temperature can be assumed as $R = 1\Omega \times (1 + \theta\, \Delta T_{degC})$ where θ is the thermal coefficient of the resistor and ΔT_{degC} is the extra temperature on top of 25 degC.

Select an option

Clear Response

- 0.14 ohm/degC
- 0.16 ohm/degC
- 0.18 ohm/degC
- 0.20 ohm/degC
- 0.24 ohm/degC

Next

Question 27

Revisit Later

Every badminton doubles match has participation of 4 players, with 2 players in one team, and the other 2 players in the other team. Consider a group of 7 players who regularly play badminton.

How many unique combination of doubles matches can happen with 7 players? For example, when players P, Q, R, S play a doubles match, the configuration "P, Q vs R, S" is same as "Q, P vs R, S" but different than "P, R vs S, Q"

Select an option

Clear Response

- 21
- 105
- 126
- 42

Next

Question 28

Revisit Later

Every badminton doubles match has participation of 4 players, with 2 players in one team, and the other 2 players in the other team. Consider a group of 7 players who regularly play badminton.

Suppose each doubles match lasts 10 mins, and the total duration for which the players have booked a badminton court is 1 hour. What is the expected value of average matches played by each player?

Select an option

Clear Response

- 4.00
- 1.71
- 3.43
- 3.82
- 2.86

Next

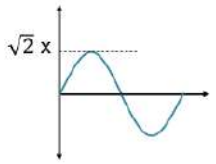
Question 29

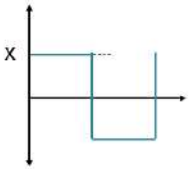
Revisit Later

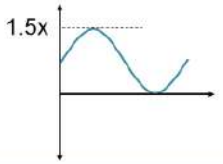
Which of the below waveforms has the highest RMS (root mean square) value? Only one period of the waveform is shown, but for the purpose of the analysis, please consider periodic waveforms running from –infinity to + infinity.

Select an option

Clear Response







Next

Question 30

Revisit Later

Suppose 2 Cr scooters are sold in India annually. Let us make a simplistic assumption that life of each two-wheeler is 10 years. Suppose the number of two-wheelers sold per year has stabilized to 2 Cr and the market is not growing. What is a good estimate for total number of functional two-wheelers in India?

Select an option

Clear Response

- 2 Cr
- 10 Cr
- 20 Cr
- 24 Cr
- Insufficient information to make any reasonable guess

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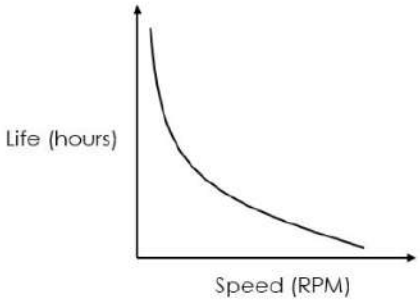
Question 31

Revisit Later

Suppose an electric motor's life has been characterized at different speeds as shown in the figure. The analytical expression for life in hours as a function of speed was found to be roughly:

Life = 8,000,000/Speed

For e.g., if the motor is operated continuously at 8000 RPM, the motor breaks down after about 1000 hours.



Suppose the motor is operated such that its speed is linearly increased from 0 RPM to indefinitely high speeds at a rate of 100 RPM per hour (i.e., after 1 hour, the motor's speed will be 100 RPM, after 2 hour, the motor's speed will be 200

Select an option

Clear Response

- 20,000 RPM
- 40,000 RPM
- 60,000 RPM
- 80,000 RPM

Next

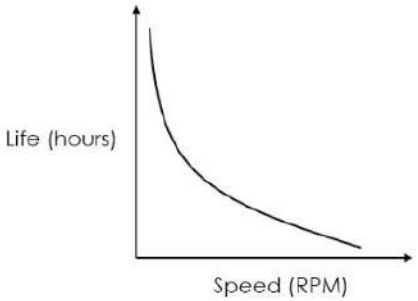
Question 32

Revisit Later

Suppose an electric motor's life has been characterized at different speeds as shown in the figure. The analytical expression for life in hours as a function of speed was found to be roughly:

Life = 8,000,000/Speed

For e.g., if the motor is operated continuously at 8000 RPM, the motor breaks down after about 1000 hours.



Suppose the motor is subjected to below drive cycle, what is the expected lifetime of the motor?

Select an option

Clear Response

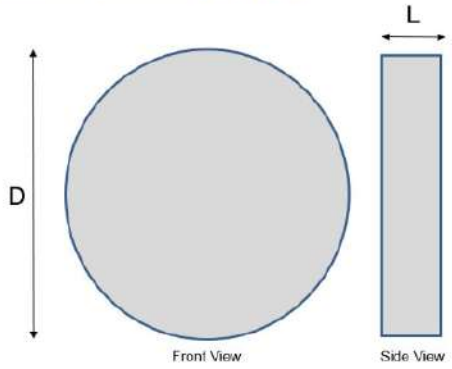
- 1600 hours
- 2048 hours
- 3200 hours
- 4096 hours

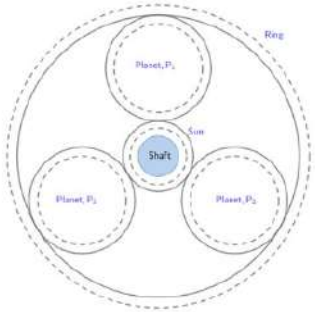
Next

Question 33

Revisit Later

Consider a cylindrical volume in space defined by diameter D = **120 mm** and axial length L = **12 mm**. It is required to design a planetary gear system which can be housed in the defined volume. The input power is provided at the sun gear and output is taken from the ring while the centers of the three planets are held fixed. Some external design constraints dictate that the shaft rotating the sun must have a minimum diameter of **20 mm**.





Schematic of a typical Planetary gear system

If the sun, planet and ring has m, n and k teeth respectively. Which of the following relations hold true ?

Select an option

Clear Response

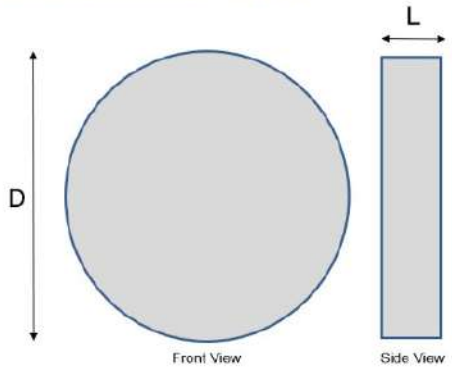
- **m + n = k**
- **m + 2n = k**
- **2m + n = k**
- **2m = k + n**

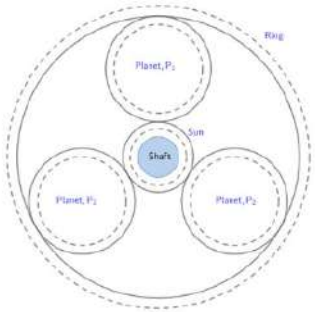
Next

Question 34

Revisit Later

Consider a cylindrical volume in space defined by diameter D = **120 mm** and axial length L = **12 mm**. It is required to design a planetary gear system which can be housed in the defined volume. The input power is provided at the sun gear and output is taken from the ring while the centers of the three planets are held fixed. Some external design constraints dictate that the shaft rotating the sun must have a minimum diameter of **20 mm**.





Schematic of a typical Planetary gear system

Which amongst the following gear teeth combinations for sun, planet and ring respectively provide the maximum speed reduction ratio, while respecting the constraints mentioned ?

Select an option

Clear Response

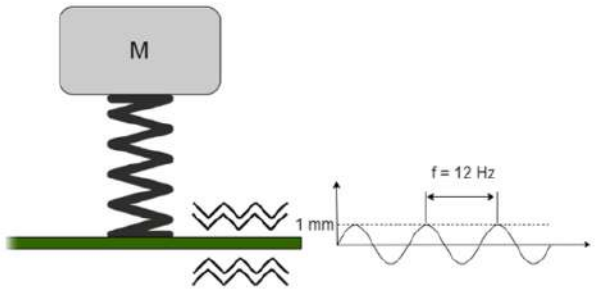
- 23, 59, 141
- 25, 58, 141
- 23, 46, 115
- 21, 50, 122

Question 35

Revisit Later

The seat of a Motor bike, with the rider weighs **M=100 Kg** and is found to have a static deflection of $\frac{250}{169}$ **mm** under the self-weight. The vibration of engine is transmitted to the base of the seat as harmonic motion with frequency **12 Hz** and amplitude **1 mm**.

{assume g= 10 m/sec^ 2 }
{assume π = sqrt(10) }
Feel free to assume lump-model as shown



What is the level of vibration felt by the rider?

Select an option

Clear Response

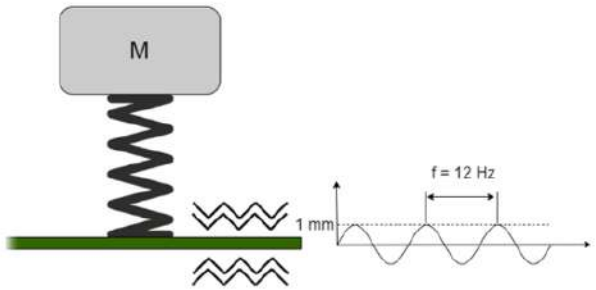
- 6.76 mm at 12 Hz
- 8.45 mm at 13 Hz
- 6.76 mm at 13 Hz
- 8.45 mm at 12 Hz

Question 36

Revisit Later

The seat of a Motor bike, with the rider weighs **M=100 Kg** and is found to have a static deflection of $\frac{250}{169}$ **mm** under the self-weight. The vibration of engine is transmitted to the base of the seat as harmonic motion with frequency **12 Hz** and amplitude **1 mm**.

{assume g= 10 m/sec^ 2 }
{assume π = sqrt(10) }
Feel free to assume lump-model as shown



Which of the following will guaranty in reducing the effect of vibration?

Select an option

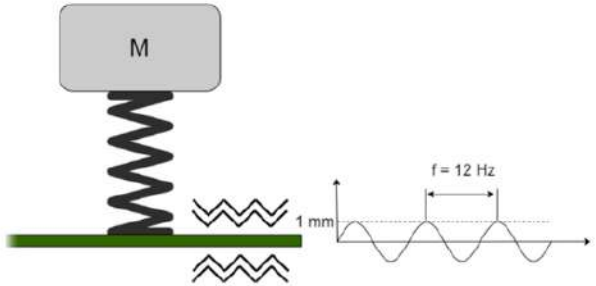
Clear Response

- Both 1 and 4
- Both 2 and 3
- Both 1 and 2
- None of statements

Next

transmitted to the base of the seat as harmonic motion with frequency **12 Hz** and amplitude **1 mm**.

{assume g= 10 m/sec^ 2 }
{assume π = sqrt(10) }
Feel free to assume lump-model as shown



Which of the following will guaranty in reducing the effect of vibration?

1. Increase stiffness of the seat
2. Reduce weight of the seat
3. Decrease stiffness of the seat
4. Increase weight of the seat

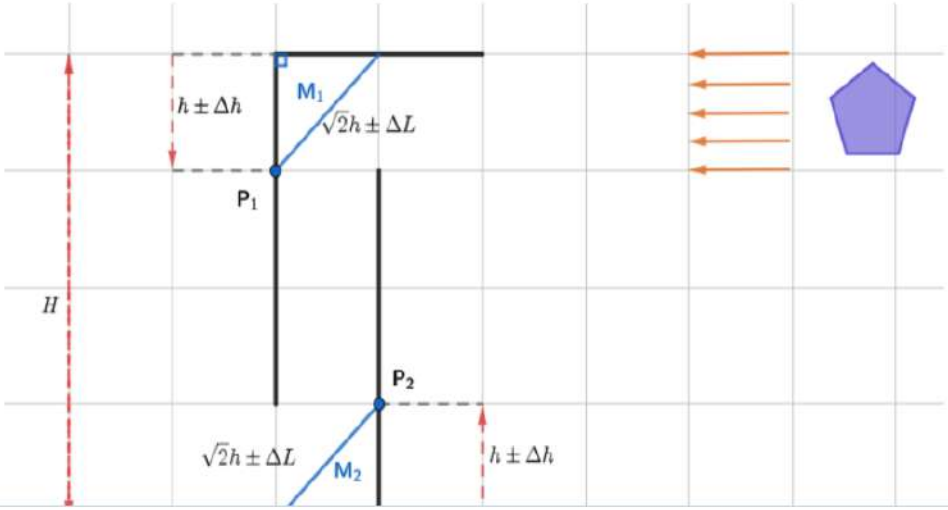
Select an option Clear Response

- Both 1 and 4
- Both 2 and 3
- Both 1 and 2
- None of statements

Question 37

Revisit Later

Consider the following periscope-like system, consisting of two plane mirrors M_1 and M_2. The base length of the mirrors along with the tolerance in them, is mentioned in the figure. The points where the two mirrors are mounted on the vertical wall is also shown with the tolerance in their location. Assume that both the vertical walls are parallel, and all other features of the system are exact. An illuminated object is kept at infinity in front of the periscope.



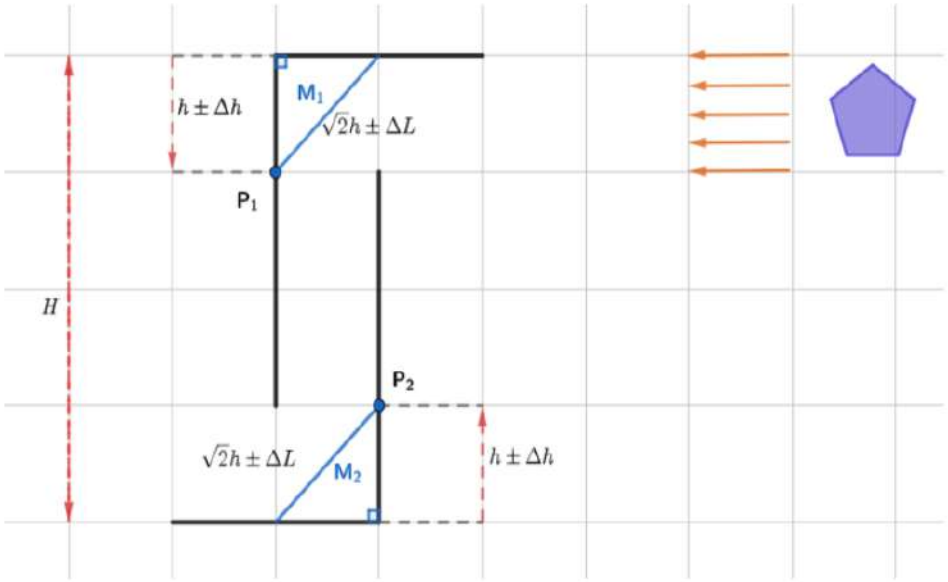
Select an option

Clear Response

- $\frac{\pi}{4} \pm \frac{1}{\sqrt{2}h}\left(\sqrt{2}\Delta h + \Delta L\right)$
- $\frac{\pi}{4} \pm \frac{1}{\sqrt{2}h}\left(\sqrt{2}\Delta h - \Delta L\right)$
- $\frac{\pi}{4} \pm \frac{1}{\sqrt{2}h}(\Delta h + \Delta L)$
- $\frac{\pi}{4} \pm \frac{1}{h}\left(\sqrt{2}\Delta h + \Delta L\right)$

Next

parallel, and all other features of the system are exact. An illuminated object is kept at infinity in front of the periscope.



Choose the correct description of the angle(in rad) made by the mirrors with the horizontal wall?
[For calculation assume that the tolerance values are very small in comparison to the base dimensions]

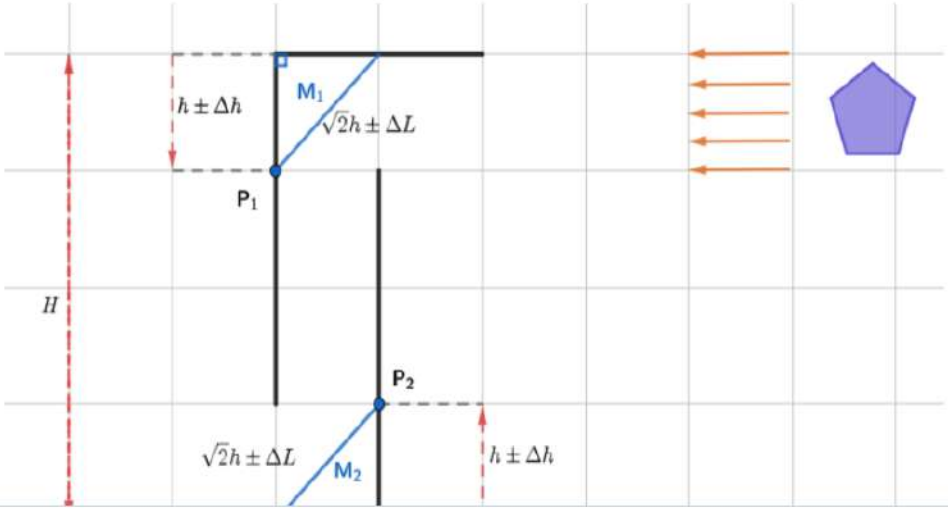
Select an option | Clear Response

- $\frac{\pi}{4} \pm \frac{1}{\sqrt{2}h}\left(\sqrt{2}\Delta h + \Delta L\right)$
- $\frac{\pi}{4} \pm \frac{1}{\sqrt{2}h}\left(\sqrt{2}\Delta h - \Delta L\right)$
- $\frac{\pi}{4} \pm \frac{1}{\sqrt{2}h}(\Delta h + \Delta L)$
- $\frac{\pi}{4} \pm \frac{1}{h}\left(\sqrt{2}\Delta h + \Delta L\right)$

Question 38

Revisit Later

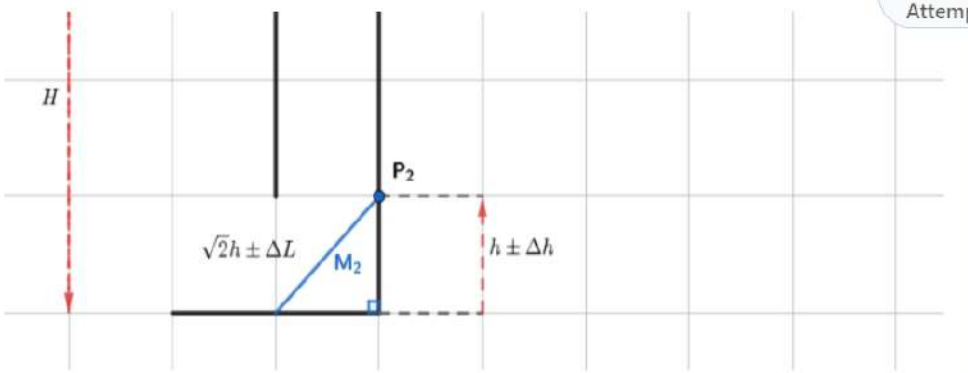
Consider the following periscope-like system, consisting of two plane mirrors M_1 and M_2. The base length of the mirrors along with the tolerance in them, is mentioned in the figure. The points where the two mirrors are mounted on the vertical wall is also shown with the tolerance in their location. Assume that both the vertical walls are parallel, and all other features of the system are exact. An illuminated object is kept at infinity in front of the periscope.



Select an option

Clear Response

- 2 & 4
- Only 3
- 1 & 3
- 5 & 6
- Only 6



What are the worst-case scenarios in which, **if** it is verified that the object is visible on the other side, then it would be ensured that the object is visible in all possible configurations specified by the tolerance ranges ?
[Example scenario:: M_1 : smallest ; P1: highest → means M_1 is of smallest possible length and P_1 is at highest possible location]

1. M_1 : smallest, P_1: lowest ; M_2 : largest, P_2 : lowest
2. M_1 : smallest, P_1: lowest ; M_2 : smallest, P_2: highest
3. M_1 : largest, P_1: highest ; M_2 : smallest, P_2 : highest
4. M_1 : smallest, P_1: highest ; M_2 : smallest, P_2: lowest
5. M_1 : smallest, P_1: lowest ; M_2 : largest, P_2: highest
6. M_1 : largest, P_1: highest ; M_2 : smallest, P_2: lowest

Select an option | Clear Response

- ○ 2 & 4
- ○ Only 3
- ○ 1 & 3
- ○ 5 & 6
- ○ Only 6